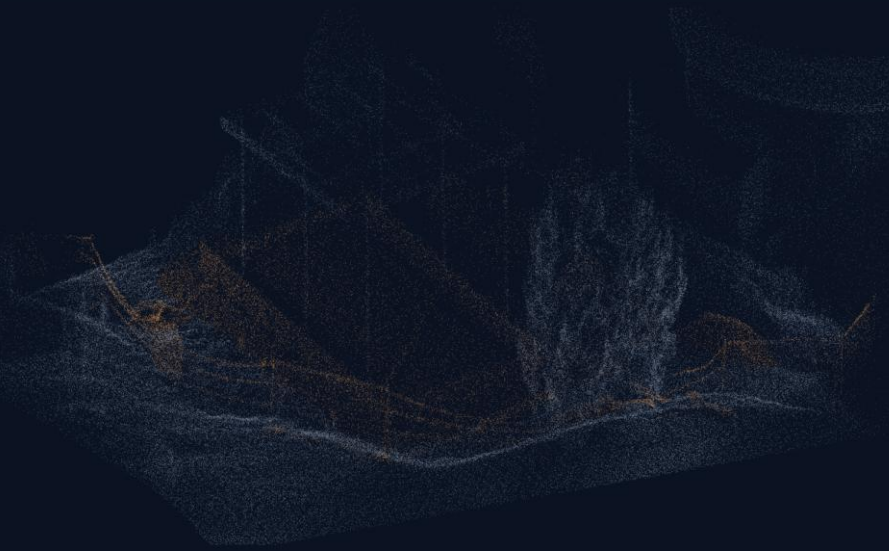
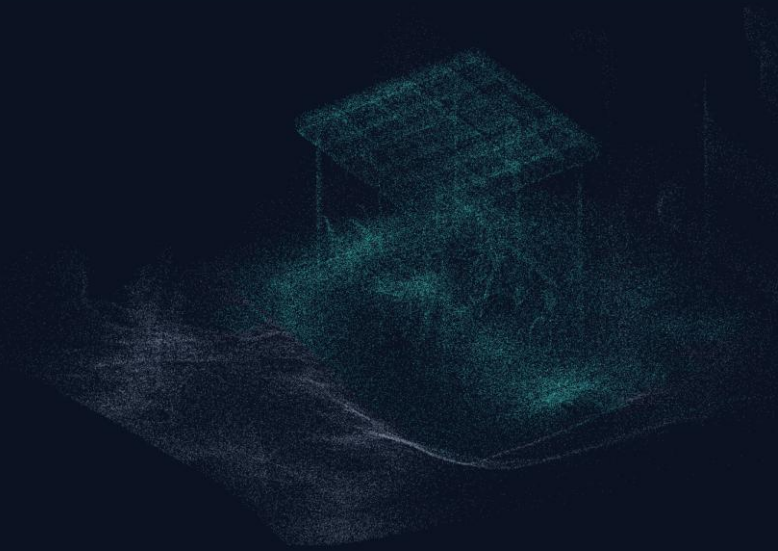
Can AI distinguish real construction progress from 3D scanning errors?



EARLIER · EPOCH 0



LATER · EPOCH 3

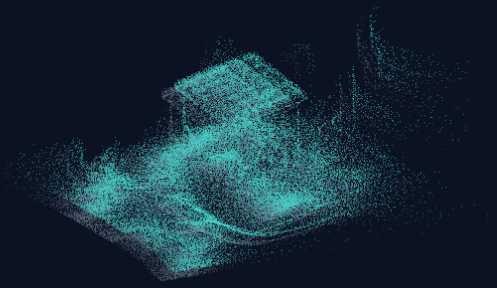
Robust 3D Change Detection for Construction Progress Monitoring

Vishwanath Venugopal · CE23B124 / Shaik Koduru Asif · CE23B114

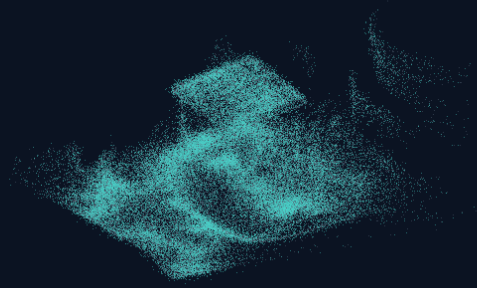
Source: MultiChange3D BikeParkingConstruction · pair 03 · genuine display samples

A difference in points is not always a change in work.

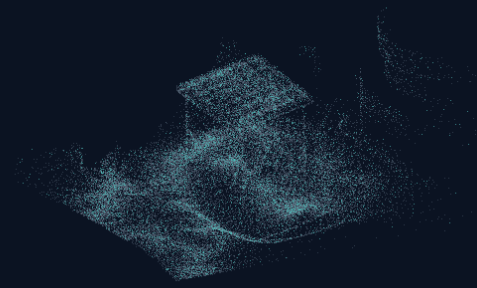
Misregistration



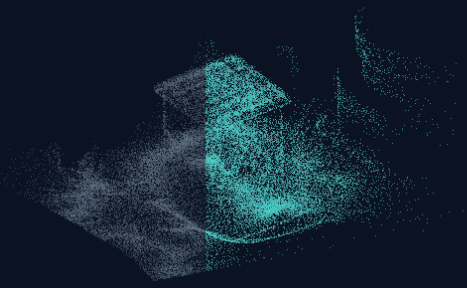
Noise



Sparsity



Missing view



Same surface. Different observations.

Displacement ≠ construction • Absence ≠ removal

Source: local pair 03 sample · explanatory perturbations, not benchmark measurements

Learn local evidence. Test what survives.

Can multiscale learned matching outperform geometric distance on sampled construction cores?

01

Benchmark: added / removed / unchanged

02

Robustness: simulate scanning errors

03

Review: inspect errors and candidate regions

One site / same-site comparisons / human judgement remains essential

Sources: coverage design.json · interpretation_scope.json · original methods

Four epochs. Six pairs. Two directions.

EPOCH 0 | EPOCH 1 | EPOCH 2 | EPOCH 3

01 02 03 12 13 23

MultiChange3D · one construction site · point labels: changed / unchanged

Earlier → later

Candidate removals on earlier points

Later → earlier

Candidate additions on later points

Source: BikeParkingConstruction · coverage design.json · fold_0/protocol.json

From scans to decisions—with evidence attached.

Paired scans › Buffered cores › Local evidence › Change scores › Review regions

Geometry baseline

Nearest distance

Surface-normal projection

Learned comparison

Paired multiscale patches

+ bounded geometry features

Output contract

Sampled point evidence

—not a full-cloud result

Sources: multipair_data.py · learning_v2_data.py · learning_v2_model.py · construction_change_regions.py

Three ways to interpret a missing neighbour.

C2C

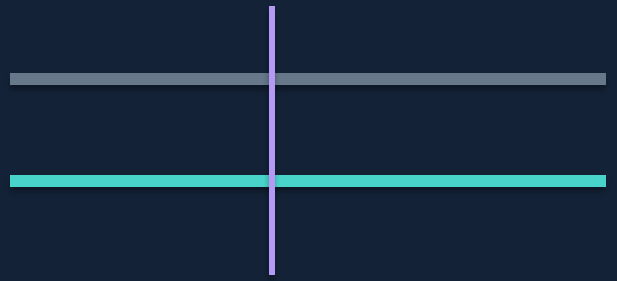


↕ nearest distance

Distance > threshold

Fixed reference: 0.075 m

M3C2-style



Project along a normal

Local support + uncertainty

Unknown when unsupported

Multiscale V2



0.15 / 0.40 / 0.80 m

Shared patch encoder

Weighted BCE—not focal loss

Sources: c2c_baseline.py · m3c2_baseline.py · learning_v2_model.py · coverage design.json

Spatial separation—not random point splitting.



8 m

Grid cells

2.5 m

Core boundary guard

3,000 train / 1,200 validation / 1,500 test

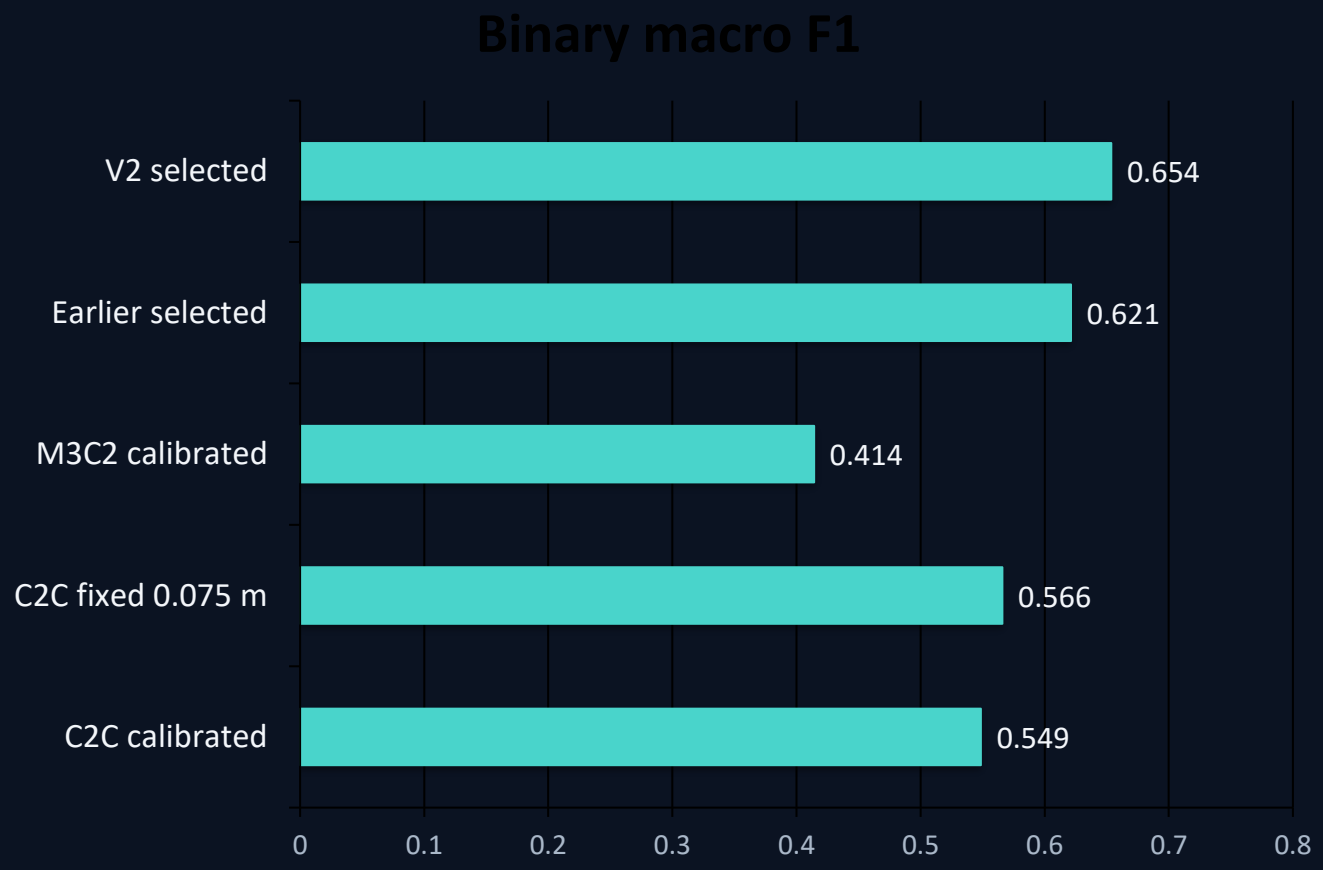
Counts per scan-direction, pair and fold

90,000 clean test observations · 5 folds

5 m cross-split guard; one site, reused validation

Source: coverage fold_0/protocol.json · design.json · interpretation_scope.json

V2 improves clean comparison. Changed F1 stays modest.



0.654

Mean binary macro F1

0.390

Mean directional changed F1

+10.51 pp vs calibrated C2C

+8.78 pp vs fixed C2C · F1, not accuracy

Source: coverage analysis/clean_comparison.csv · 30 pair-fold evaluations · uniform sampled test cores

One model. Different questions. Different F1s.

0.654
Mean binary macro F1

0.390
Mean changed F1

0.741
Pooled 3-class macro F1

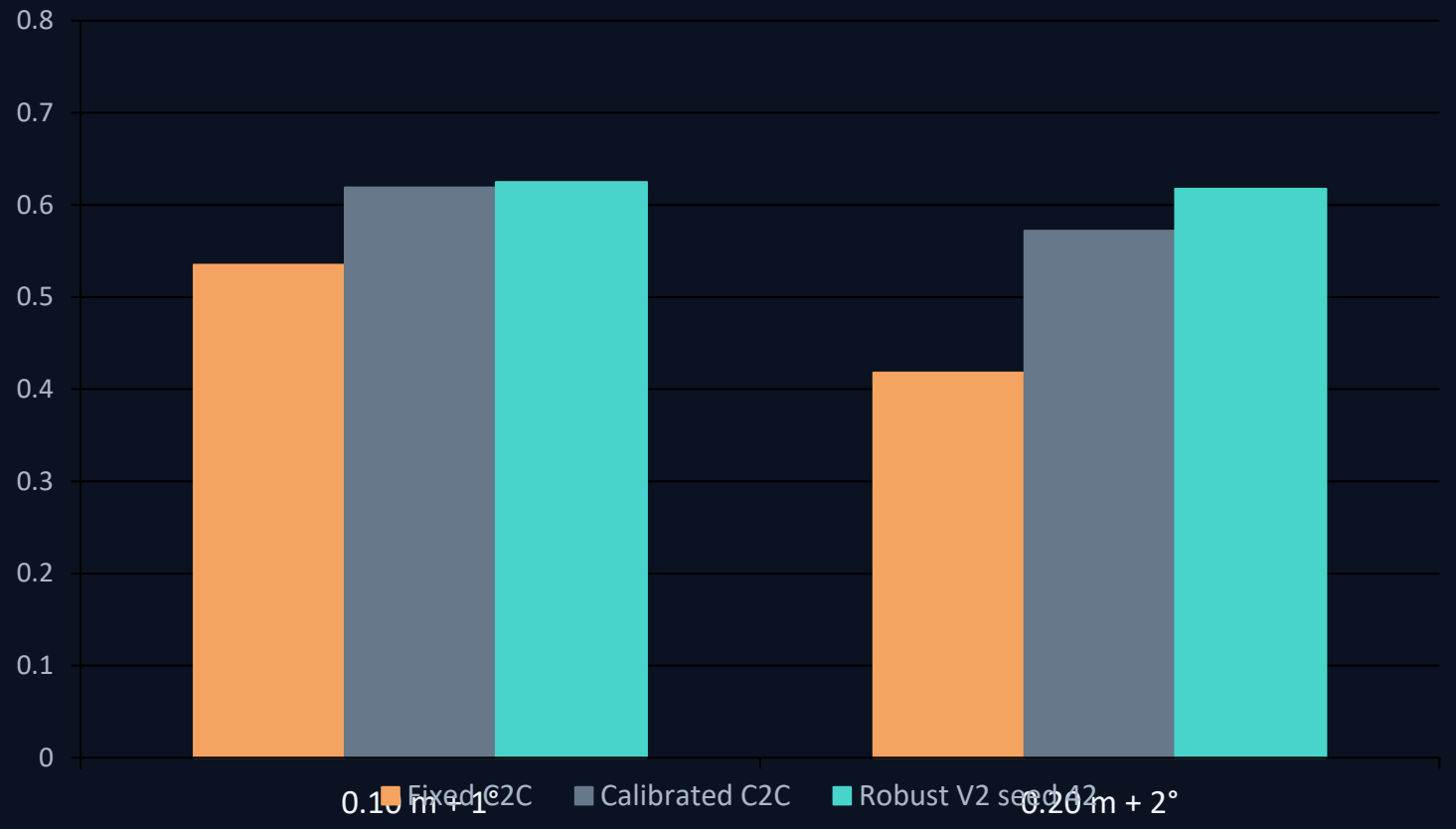
21 of 60 directions contain no labelled change.

Absent-class F1 is set to zero.

Perfect mean binary macro F1 is therefore 0.825—not 1.0.

Sources: class_support_and_pooled_confusions.csv · metric_ceiling_diagnostic.json

Robust training helps under controlled misalignment.



0.618

Severe combined · robust V2

0.418

Severe combined · fixed C2C

Translation + rotation · no added noise

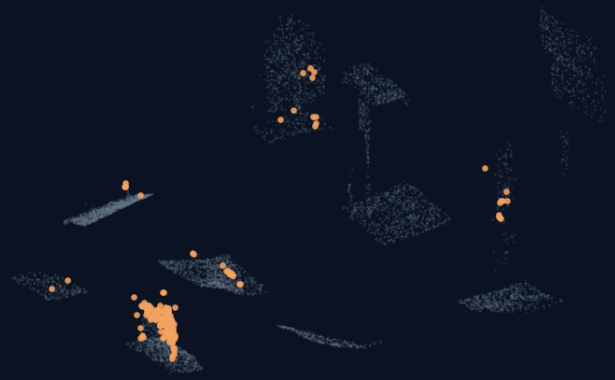
Calibrated C2C: 0.572 · robust V2: 0.618

Fold 0 only; not selected five-fold workflow

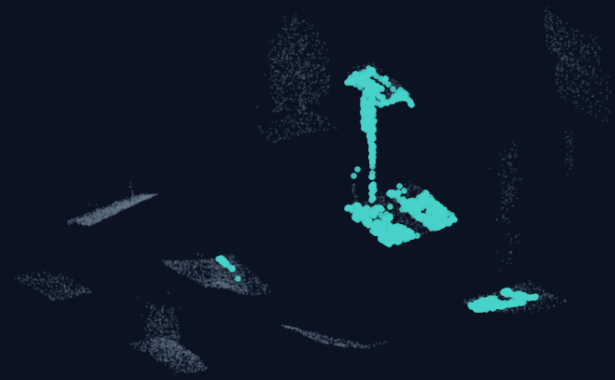
Source: stress_condition_summary.csv · combined conditions · six pairs × three perturbation seeds

Expose the errors—not just the winning score.

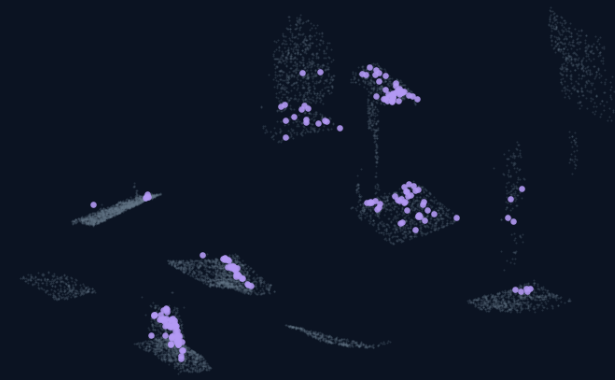
False alarms (FP)
169 displayed cores



Missed change (FN)
886 displayed cores



Near decision boundary
188 displayed cores

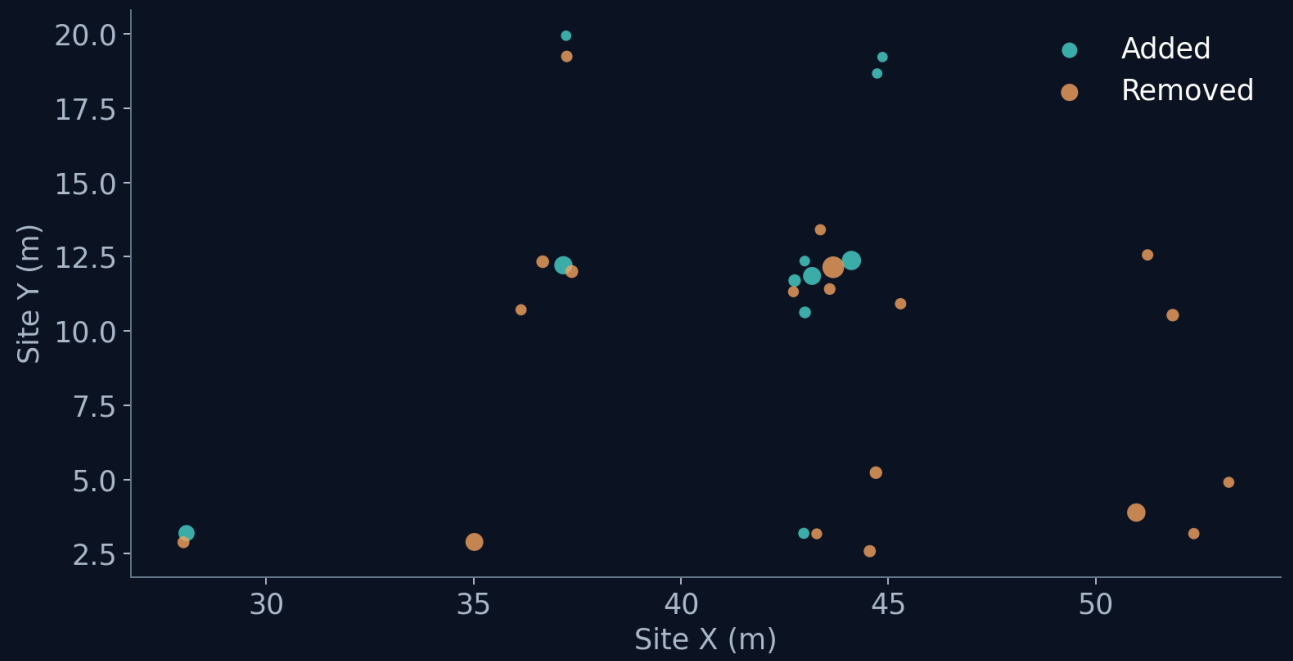


FP: review a false alarm / FN: recover missed evidence

Purple marks |score − 0.5| < 0.1, not a calibrated uncertainty probability.

Source: construction_change_regions pair_03_added_cores.npz · all 7,500 later-epoch sampled out-of-fold cores

191 candidate regions—not 191 construction objects.



191

Across six pairwise comparisons

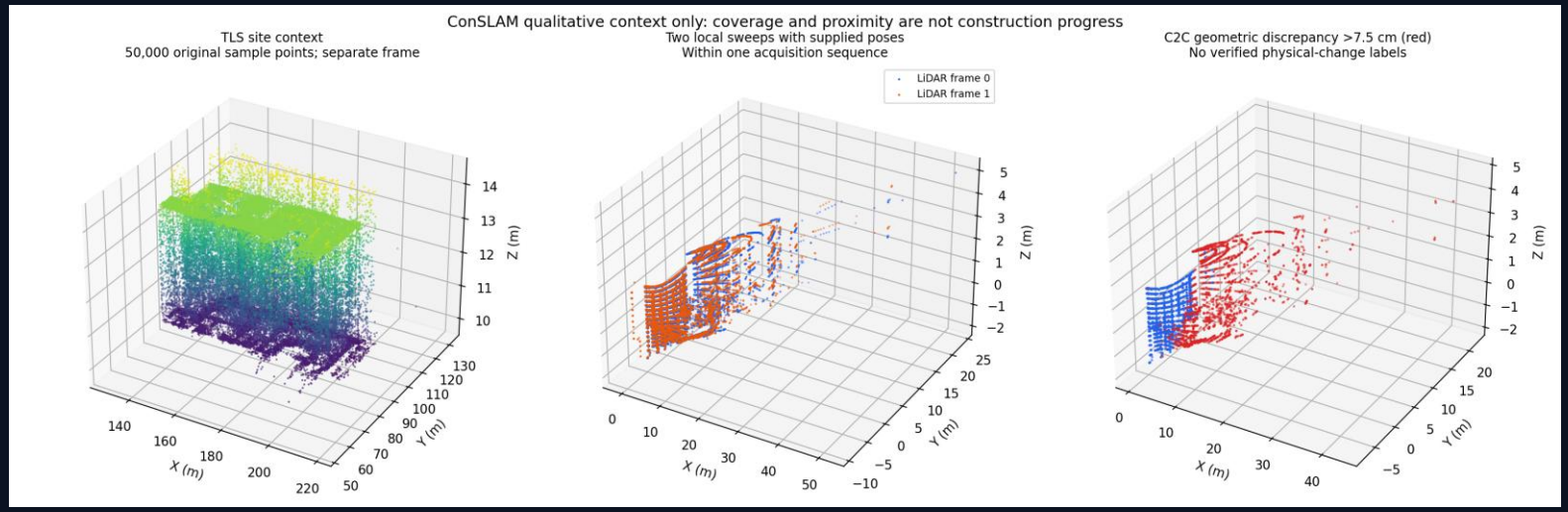
83

Only unchanged-labelled sampled members

0.40 m grouping radius / ≥3 positive sampled cores

Sources: candidate_regions.csv · pair_summary.csv · design.json · verification.json

ConSLAM is a reality check—not a labelled test.



4.84 seconds

Between selected mobile frames

No official change labels.

No supervised V2 inference.

No F1 or IoU claim.

Source: conslam_qualitative_2026-10-25_v2/inspection.json and conslam_qualitative.png

Detect evidence. Review context. Verify progress.

IMPLEMENTED

Sampled change scores

Error inspection + candidate grouping

FUTURE

Verified BIM association

Quantity / schedule integration

V2 improves same-site evidence. It does not certify installed work.

NEXT 5-minute offline evidence demonstration

Sources: clean_comparison.csv · construction candidate design.json · qualitative inspection.json

Exact metrics and the absent-class ceiling.

METHOD	BINARY MACRO	CHANGED F1	UNKNOWN
C2C calibrated	0.548563	0.241073	0.00%
C2C fixed 0.075 m	0.565899	0.299985	0.00%
M3C2 calibrated	0.414228	0.034837	18.67%
Earlier selected	0.621191	0.345500	0.00%
V2 selected	0.653659	0.389989	0.00%

Pooled 3-class macro: 0.740508 | Oracle mean binary ceiling: 0.825

39 positive-support + 21 unchanged-only directions; zero convention retained.

Sources: clean_comparison.csv · class_support_and_pooled_confusions.csv · metric_ceiling_diagnostic.json

V2 is a custom metric-fusion network.

3 paired scales › Shared 8→32→64 › Max + mean › Interactions › Geometry fusion

Patch branch: 1,536 → 128 / Geometry branch: 165 → 64

Concatenate 192 → 64 → 1 logit

Checkpoint: minimum clean-validation BCE

Threshold + variant: clean-validation binary macro F1

Loss: equal pair / present-class weighted BCE

Sources: learning_v2_model.py · coverage design.json · interpretation_scope.json

Follow-ups are diagnostics—not extra headline wins.

Do not promote an exploratory improvement without validation and matching scope.

Local follow-ups

Intensity / temporal / fusion

Not the headline test claim

Open_space

Separate exposed indoor diagnostic

C2C 0.860 / selected V2 0.786

What is still missing

Multiple construction sites

Verified component semantics

Sources: local follow-up manifests · external_open_space/metrics.csv · interpretation_scope.json